Yu-Hsuan Tsai

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EDUCATION

National Taiwan University Taipei, Taiwan
M.S. in Computer Science and Information Engineering Sep 2026 – Expected 2028
Web Mining and Information Retrieval Lab; Advisor: Prof. Pu-Jen Cheng.
Research interests: Information retrieval and machine learning.

National Yang Ming Chiao Tung University Hsinchu, Taiwan
B.S. in Computer Science Sep 2022 – Jun 2026
Relevant Coursework: Data Structures & OOP (A), Algorithms (A),
Artificial Intelligence (A+), Software Engineering (A+).

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript, JavaScript, C#, Swift.
Software: React, Next.js, Node.js, Electron, PySide6, WebSocket, Git, Unity.
ML / Data: PyTorch, YOLO, OpenCV, NumPy, pandas; image and signal processing.

EXPERIENCE

Logitech
Software Intern, Mechatronics Team Mar 2026 – Sep 2026

- Developed a Python desktop tool for keyboard backlight evaluation, combining measurement processing, YOLO-based key detection, image matching, and automated test reporting.
- Built TypeScript/React interfaces and Node.js WebSocket services for peripheral configuration, calibration, and live telemetry; integrated device communication and debugged firmware/SDK issues.
- Developed a Python input-recognition tool with a desktop interface, connecting input capture, gesture recognition, and configurable system actions.

National Taiwan University
Teaching Assistant, Systems Programming Sep 2026 – Present

- Design systems programming assignments and automated tests to validate program correctness and edge-case behavior.

Cognitive Engineering Lab
Research Assistant 2024 – 2025

- Built Python pipelines for EEG preprocessing and feature extraction; supported data flow between hardware sensors and real-time analysis workflows.

PROJECTS

LiveSubtitle — iOS Subtitle and Translation App 2026

- Developed a Swift iOS app integrating native video playback, Japanese speech recognition, and on-device translation with time-aligned subtitle rendering.
- Implemented asynchronous audio processing, subtitle caching, and cancellation of stale work after playback seeks; integrated local language-model inference through llama.cpp.

XR Interaction Projects — Unity / C# / Meta Quest 2025

- Built two interactive experiences: a two-player asymmetric puzzle and an environment-adaptive game, implementing physical–virtual interaction and gesture-based controls.
- Projects received Bronze and Special Presentation Awards at the NYCU XR Showcase.